R-AI-D (Rtificial Audio Intelligence Distribution)

Rtificially intelligent audio for Linux

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Acknowledgements

Project Bluefin

R-AI-D would not exist without the development environment provided by Project Bluefin. An atomic GNOME operating system plus dinosaurs plus cloud-native tools plus the NVIDIA Container Toolkit plus dinosaurs plus Flathub plus Homebrew. And dinosaurs. *Deinonychus, Achillobator, Dakotaraptor* - take your pick.

Ubuntu 26.04 LTS "Resolute Raccoon"

Why Ubuntu LTS? Two reasons:

1. It's popular - the Ubuntu team made a conscious effort from the beginning to make a Linux distribution for everyone.
2. As a result of that popularity, nearly all of the third-party software that runs on Linux is tested on Ubuntu LTS, and some projects with limited resources can only afford to test on Ubuntu LTS.

The R ecosystem

R

I've been programming in R since 2000. As a scientific applications programmer, it's closer to how I think than any other mainstream language. It's the perfect blend of imperative, functional and object-oriented programming paradigms.

RStudio Server, open source edition

i'm not a big fan of IDEs in general; I usually work best at the command line where I can automate as much as possible. But for R literate programming / package development, I've been using RStudio since it was in beta. The open source edition of RStudio Server is in the R-AI-D container by default.

Rocker, r2u, rstudio2u and BSPM

Previous versions of R-AI-D used `r2u` from the Rocker project and `rstudio2u` from Professor Jeffrey Girard as base images. I very recently switched to building the image from the base Docker Hub Ubuntu image, using `bspm` by Professor Iñaki Ucar.

The primary reason I switched to direct builds from the components is that I want to make it easy to "port" R-AI-D from a container to a conventional Linux desktop. `bspm` supports Fedora, openSUSE and Arch Linux in addition to Ubuntu.

Posit skills

Posit skills are the core open-source component of Posit's recently-announced Posit Assistant. Posit Assistant works with the version of RStudio Server in the R-AI-D container - I've tested it. But I can't re-distribute it.

Ollama

There are a number of open source runtimes for local large language model operations, but Ollama is the simplest and seems to be the most popular.

This is a Quarto book.

To learn more about Quarto books visit https://quarto.org/docs/books.

1 Introduction

1.1 What is this?

Rtificial Audio Intelligence Distribution (R-AI-D) is a collection of software that creates a Podman (Walsh 2023) container for audio processing in R. The resulting container features:

- the Ubuntu 26.04 LTS "Resolute Raccoon" operating system,
- the R programming language (R Core Team 2022),
- r2u: CRAN as Ubuntu Binaries
- bspm: Bridge to System Package Manager (Ucar 2026),
- RStudio Server open source edition,
- the Quarto scientific and technical publishing system,
- R packages for package development (Wickham and Bryan 2023),
- R packages for interfacing with AI tools (Verde Arregoitia, Luis D. 2026):
  - ellmer (Wickham et al. 2025),
  - mcptools (Couch et al. 2026),
  - ollamar (Lin and Safi 2025),
  - ragnar (Kalinowski and Falbel 2026),
  - shinychat (Cheng et al. 2025),
  - vitals (Couch 2025),
- R packages for audio analysis and synthesis (J. Sueur et al. 2008), and
- the Ollama framework for managing local models.

R-AI-D was developed on Bluefin DX, but should run on any `x86_64` Linux host system supporting Podman. If an NVIDIA GPU is present, the R-AI-D build process will detect it and Ollama will use it.

Like its Bluefix DX inspiration, the R-AI-D command line features the Starship cross-shell prompt generator. The Cascaydia Cove nerd font is included.

1.2 Licensing, contributing, roadmap, etc.

R-AI-D uses the Creative Commons CC0 license, which is a bunch of words that say mostly "R-AI-D is public domain." R-AI-D is mostly `bash` scripts that download some open source software and build it into a container. Why would I copyright that?

Contributing? Forks, bug reports and feature requests are welcome. Pull requests, on the other hand, probably not. I don't have the time to review other peoples' code beyond fixing typos.

The roadmap is mostly determined by what I need and when. I am building an R package for assisting composers of xentonal music into R-AI-D, and that may well add requirements to what's in the container now.

2 Quick Start

2.1 Hosting

The Ollama AI framework works on many hosts, but the only hardware I have avaiable for testing is a laptop with an NVIDIA GeForce GTX 1650 Ti and a desktop with an NVIDIA GeForce RTX 3090. I develop on Bluefin, which is a dinosaur-themed expansion of Fedora Silverblue that has NVIDIA support built in.

For other hosts, you will need Podman 4.9.3 or later. If you have an NVIDIA GPU ("Turing" or later), you will need the NVIDIA drivers and NVIDIA Container Toolkit installed and configured.

2.2 Building the container

1. Create the `SHARED_WITH_CONTAINER` directory in your home directory, `cd` into it, and clone the `R-AI-D` repository.

   ```
   mkdir --parents $HOME/SHARED_WITH_CONTAINER
   cd $HOME/SHARED_WITH_CONTAINER
   git clone --recurse-submodules https://github.com/AlgoCompSynth/R-AI-
   D.git
   ```

2. Build the image.

   ```
   cd R-AI-D/container-creation/
   ./1_build_image.sh
   ```

3. Create and run the container.

   ```
   ./2_run_container.sh
   ```

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